Attitude and Size Estimation of Satellite Targets Based on ISAR Image Interpretation

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***Abstract*— The attitude and size of satellite targets are essential information for their activity analysis. This article proposes a novel approach to estimate the absolute attitude and size of satellite targets in the 3-D stable coordinates based on inverse synthetic aperture radar (ISAR) image interpretation. In an ISAR image of a satellite, the satellite's body is chosen as an individual structure segmented from each ISAR image by employing pix2pix generative adversarial network (Pix2pixGAN). By exploring the shape feature of the satellite body with principal component analysis (PCA), the satellite attitude and size are estimated jointly through solving an optimization based on the gradient iteration method. The optimization is established by bridging range-Doppler (RD) images and the target feature parameters (attitude and size) with the accommodation of target trajectory information and the ISAR geometric projection model. In the experiments, the simulation data are generated from real satellite orbital parameters and the computer-aided-design (CAD) models of two satellite targets: TianGong (TG) and KeyHole (KH). Compared with the factorization-based reconstruction method, the proposed method can estimate the attitude and size of the satellite simultaneously and has a higher size estimation accuracy.**

***Index Terms*— Attitude estimation, geometry characteristics, inverse synthetic aperture radar (ISAR), pix2pix generative adversarial network (Pix2pixGAN), principal component analysis (PCA), satellite targets.**

I. Introduction

With increasing numbers of space targets (satellite, spacecraft, debris, and so on) being launched into space, it becomes essential and challenging to analyze satellites' state and geometry characteristics due to the increasingly complex and crowded space environment [1], [2]. An important aspect of state analysis of satellite target is to determinate the attitude, which has a large domain of applications, such as the salvage of faulty satellite, determination of spy satellites' state, or its military activities [3]–[7]. Another aspect of state analysis is to obtain satellite size, which is an important feature for satellite classification and recognition. Due to the wide application in civilian and military fields and the capability of providing 2-D high-resolution images of moving targets [8]–[10], radar is generally adopted to accomplish state analysis tasks. Furthermore, due to the abundant structure information contained in high-resolution inverse synthetic aperture radar (ISAR) images, one can obtain 3-D geometry characteristics of satellites from typical structural features of targets in radar images (e.g., satellite body and satellite solar panels), combined with the satellite's motion characteristics [11].

Several works have been developed to acquire satellite state information, including satellite attitude and geometry characteristics, and they can generally be sorted into three categories. One basic approach uses various attitude sensors, including gyroscopes [12], sun sensors, star sensors, and magnetometers, to record the 3-D rotation angle variation of a satellite. Then, these rotation angles are filtered by Kalman filter (KF) [13], [14], extended Kalman filter (EKF) [15], or unscented Kalman filter (UKF) [16], [17]. Finally, a satellite attitude can be obtained. These algorithms are suitable for spacecraft equipped with three-axis gyros and attitude sensors, which are impracticable in most practical applications due to the noncooperative of most space targets. The second category relies on the preexisting simulation database, which establishes a complete full-view simulation database by electromagnetic simulation techniques. The measured data, including the radar cross section (RCS), the high-resolution range profile (HRRP), and the ISAR imagery, can be used to match with the well-simulated database [6], [18]. Finally, the attitude parameters can be solved by using some optimal search algorithms. Although this data-driven method performs well in some particular tasks, the main disadvantage of these algorithms is that they need a complete full-view simulation database. Regardless of the enormous computing time, the algorithms will fail when dealing with noncooperative targets due to the lack of a targets model. The last category acquires the target attitude by reconstructing the target 3-D point model from a multiperspective image sequence [19]–[21]. This kind of algorithms first establishes the trajectory matrix by associating scatterers in multiview images. Singular value decomposition (SVD) is usually adopted to accomplish the structural reconstruction by decomposing the trajectory matrix. For this category, the feature association of multiview images is never easy, especially when the angular glint phenomenon occurs.

In recent years, some other novel approaches have been proposed to estimate satellite attitude by employing specific

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geometry structures of satellite in ISAR images, such as the typical rectangular structures of solar panels [11], [22], [23]. Zhou *et al.* [11], [24], take advantage of the line features that are the typical characteristics of rectangular solar panels in the ISAR image sequence. A satellite's attitude can be estimated by associating different lines extracted in sequence images and solving optimization, which is an intelligent and practical way to explore satellite attitude in most situations, as most satellites are equipped with rectangular solar panels. However, the estimation precision of these methods depends mainly on the accuracy extraction of line features, which requires manuals in the actual application. What is more, some other parts of the satellite may also contain line structures, such as the cube or cylinder body of the satellite, which would also influence the line feature extraction. In [22], the solar panel is treated as a plane whose normal vector can be estimated by solving ISAR images' autofocus optimization. During the autofocus process, the spatial-variant quadratic phase coefficients, together with attitude parameters, are determined. That is the first time that the quadratic phase error has been treated as positive information to estimate satellite attitude. The disadvantage of this algorithm is that its estimate accuracy is easily influenced by the solar panel segmentation and the signal decomposition under low SNR. Although, finally, all the algorithms mentioned above take advantage of the rectangular structure of the solar panel, this kind of structure cannot be observed under particular observation views when the solar panel is almost projected as a straight line. In this regard, the solar panel may not be a good structural choice for estimating a satellite's attitude.

Inspired by Zhou *et al.* [11], this article proposes a more robust algorithm to estimate satellite attitude and size jointly. The body of a satellite target is selected as a better structure in this article to estimate the satellite attitude and size. The projection matrix is first established to bridge the 3-D absolute attitude and the 2-D projected orientation of the satellite body in an ISAR image. The satellite body is then split from the original ISAR image by adopting an image-to-image translation method, pix2pix generative adversarial networks (Pix2PixGAN) [24]. Then, the orthogonal matching pursuit (OMP) algorithm [25] is employed to extract the scatterers of the satellite body in the segmented ISAR image. The 2-D position of these points undergoes principal component analysis (PCA) to bring out the principle eigenvector corresponding to the largest eigenvalue. These principle eigenvectors are treated as the 2-D poses of the satellite's capsules in ISAR images, bringing out the satellite's projected length. Finally, the satellite body's 3-D absolute attitude and length can be estimated by solving an optimization via a gradient-based method. Compared with the factorization-based reconstruction method, the proposed algorithm can jointly estimate the target attitude and size parameters with higher accuracy. The innovations of this article are listed as follows.

1) Compared with the factorization-based reconstruction method, the proposed method can estimate the satellite attitude and size simultaneously. The factorization method obtains the 3-D point distribution with an uncertain 3-D rotation, which leads to an unknown attitude. Thus, the factorization method can only estimate the absolute size for satellite targets. The experiments show that the size estimation accuracy of the proposed method is better than that of the factorization-based method.
2) In this article, we propose a novel ISAR image segmentation method based on Pix2pixGAN. Using the segmentation method, the satellite's capsules, which are further used to estimate the absolute attitude and size of satellite targets, are segmented from the ISAR images. Compared with other segmentation methods, such as the fully convolutional network (FCN) [26], SegNet [27], and DeepLabV3+ [28], the proposed method has a higher segmentation accuracy.
3) We propose a gradient-based method to search the optimal parameters. The attitude and size parameters are estimated jointly by solving a minimum parametric optimization iteratively. Compared with particle swarm optimization (PSO), the proposed method converges faster and estimates more accurately.

The rest of the article is organized as follows. In Section II, the projection theory during ISAR imaging is presented, and the projection matrix is also introduced. In Section III, the procedure of the proposed method is described in detail. In Section IV, the experiments have been designed to show the effectiveness of the proposed method. In Section V, some conclusions are drawn.

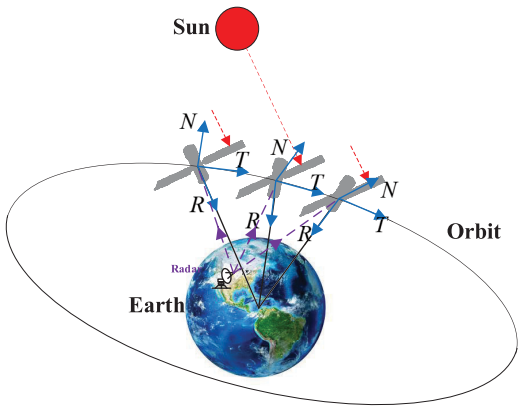


Fig. 1. Observation geometry of satellite in orbit.

II. Imaging Geometry and Projection Theory

In this section, we will introduce the radar observation geometry for satellite in orbit and the projection theory of ISAR imaging. First, the satellite is assumed to be observed in the local orbital frame or the coordinate system called RTN coordinate [18], where the coordinate system is defined by the radial (R) direction, the along-track (T) direction, and the cross-track (N) direction. This is a typical coordinate system that provides an essential reference for describing the attitude of an Earth pointing satellite [29], as shown in Fig. 1. The R-direction is commonly aligned with the direction of the satellite antenna, which is equipped on the satellite's body.

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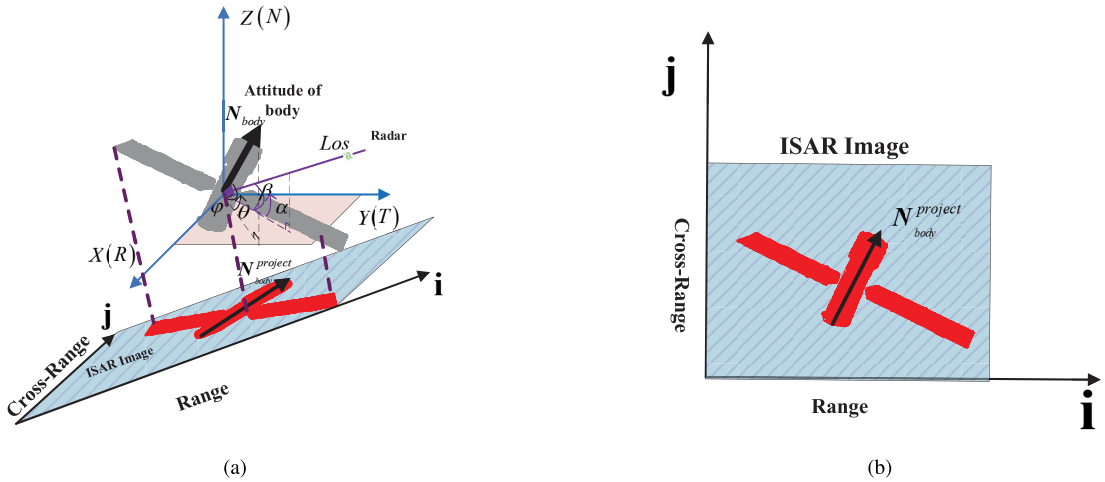


Fig. 2. ISAR imaging of satellite. (a) ISAR imaging geometry in local orbit frame and the projection process of ISAR imaging. (b) ISAR image.

To meet this condition, the satellite is kept stable in RTN coordinate under the attitude control system. It should be pointed out that the proposed algorithm applies to satellites with symmetrical capsules, such as those with cylindrical structure and cubic structure. This structural characteristic ensures that a single vector can describe the attitude of the satellite's capsules in 3-D coordinates. In practice, this condition can always be satisfied as most of the satellite's capsules are designed to be symmetrical. In the RTN reference, as shown in Fig. 2(a), the attitude of the satellite's capsule is $\mathbf{N}_{\text{body}} = [\cos\varphi \sin\theta, \cos\varphi \cos\theta, \sin\varphi]^T$, which is defined by two parameters: the azimuth angle θ and the elevation angle φ. Meanwhile, the radar line of sight (LOS) is $\mathbf{Los} = [\cos\beta(t_m)\sin\alpha(t_m), \cos\beta(t_m)\cos\alpha(t_m), \sin\beta(t_m)]^T$, which is defined by azimuth angle $\alpha(t_m)$ and the elevation angle $\beta(t_m)$, where t_m is the slow time during the observation. According to the projection theory [30], ISAR image is actually the projection of target 3-D geometry onto the 2-D imaging plane, and two orthogonal coordinate axes define the plane: the **i**-axis corresponding to the LOS direction and the **j**-axis corresponding to the Doppler direction, as shown in Fig. 2(a). According to [11], the projection vectors **i** and **j** can be expressed as (1), shown at the bottom of the page. Here, $\alpha(t_0)$ and $\beta(t_0)$ are the observation angles at time t_0; $\dot{\alpha}$, and $\dot{\beta}$ are the first derivatives of $\alpha(t_m)$ and $\beta(t_m)$; $\Delta r = c/2B$ and $\Delta f_a = \lambda/2\Omega$ are the range resolution and cross-range resolution, respectively; c is the speed of light; B is the bandwidth; λ is the wavelength; and Ω is the rotation angle of the LOS during the observation time. From (1), it can be seen that $\mathbf{i}^T\mathbf{j} = 0$, which means that the projection vectors **i** and **j** are always perpendicular to each other. With the projection vectors **i** and **j**, we can get the projection matrix expressed as

$$\mathbf{P} = \begin{bmatrix} \mathbf{i}^T \\ \mathbf{j}^T \end{bmatrix}. \tag{2}$$

As to the 3-D attitude vector of the satellite's capsule $\mathbf{N}_{\text{body}}$, it will be projected onto the ISAR imaging plane by the projection matrix, and the 2-D attitude vector $\mathbf{N}_{\text{body}}^{\text{project}}$ in the ISAR image can be expressed as

$$\mathbf{N}_{\text{body}}^{\text{project}} = \mathbf{P} \cdot \mathbf{N}_{\text{body}}. \tag{3}$$

As shown in Fig. 2(a), $\mathbf{N}_{\text{body}}^{\text{project}}$ is a unit vector that represents the 2-D orientation of the satellite's capsule. Assume that the absolute length of the satellite's capsule is L; then, the projected vector of the satellite's capsule that contains length can be expressed as

$$\mathbf{n}(\theta, \varphi; L) = L \cdot \mathbf{P} \cdot \mathbf{N}_{\text{body}}. \tag{4}$$

In the 3-D coordinates' reference, a vector can be uniquely determined by θ, φ, and L. θ and φ determine the absolute attitude of the satellite's capsule, and L determines the absolute length. To put it another way, if we get those three parameters, the satellite attitude and size can be determined. Equation (4) shows the relationship between the projected satellite's capsule vector and the absolute 3-D vector, making it possible to estimate the absolute attitude and length of the satellite from ISAR images.

So far, the imaging geometry and the projection theory of the ISAR image have been introduced. Equation (3) establishes a projection relationship between the target 3-D geometry and the 2-D images. In (3), the satellite 3-D attitude $\mathbf{N}_{\text{body}}$ is the vector that needs to be estimated in this article. On the other hand, the parameters used to build the projection matrix

$$\begin{cases} \mathbf{i} = \dfrac{[\cos\beta(t_0)\sin\alpha(t_0), \cos\beta(t_0)\cos\alpha(t_0), \sin\beta(t_0)]^T}{\Delta r} \\ \mathbf{j} = \dfrac{\left[\sin\beta(t_0)\sin\alpha(t_0)\dot{\beta} - \cos\beta(t_0)\cos\alpha(t_0)\ \dot{\alpha}, -\sin\beta(t_0)\cos\alpha(t_0)\dot{\beta} - \cos\beta(t_0)\sin\alpha(t_0)\dot{\alpha}, \cos\beta(t_0)\dot{\beta}\right]^T}{\Delta f_a} \end{cases} \tag{1}$$

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are assumed to be known beforehand according to the literature [11], which includes radar LOS, the range resolution Δr, and the cross-range resolution Δf_a. In (4), there are three parameters to be estimated: the absolute attitude of the satellite's capsule (θ, φ) and the absolute length of satellite's capsule L.

The following will establish the optimization to estimate the target absolute attitude and size with a series of 2-D ISAR images. Since the proposed method mainly focuses on solving the satellite's attitude and size estimation, some assumptions should be made beforehand. First, the satellite's well-focused ISAR images should be obtained in advance based on some autofocus methods. The resolutions of these ISAR images are assumed to be known already. Second, radar LOS $\mathbf{Los} = [\cos\beta(t_m)\sin\alpha(t_m), \cos\beta(t_m)\cos\alpha(t_m), \sin\beta(t_m)]^T$ can be calculated using radar tracking data. One can refer to [23], [31], [32] to find a detailed procedure.

III. Theory and Methodology

A. ISAR Image Segmentation

As discussed in Section II, the core idea of this article is to estimate the satellite attitude and size with the satellite's capsule part according to the series of 2-D ISAR images. First, in an ISAR image, the satellite's capsule should be segmented. There are few existing methods for ISAR image segmentation. The traditional clustering method can accomplish this task to a certain extent. This sort of method clustered different pixels according to color, brightness, and texture features. The pixels with similar characteristics are clustered into the same superpixel, and the segmented image can be obtained. However, in an ISAR image, the satellite's capsule and solar panel's pixel characteristics do not show distinct differences. It is a challenge to segment the satellite's capsule accurately. With the rapid development of deep learning networks in recent years, the existing general image segmentation methods based on deep learning have excellent performance, such as segmentation methods based on FCN [26], SegNet [27], and DeepLabV3+ [28]. For random input images, FCN trains an end-to-end full convolutional network using bilinear interpolated up-sampling, combined with layers of the feature hierarchy, refines the output's spatial precision, and realizes semantic segmentation by pixel classification. SegNet is a model with a symmetric encoding decoding structure. The decoding stage uses antipooling to recover the lost sampling information, getting smooth, and precise segmentation results. The deepLabV3+ model uses the atrous spatial pyramid pooling (ASPP) to extract the global semantic information in the coding structure and uses the image's context information to refine the segmentation results along object boundaries. However, those three methods' loss functions are generally the reconstruction loss function of the whole image, which may lose some pixel-level details. In recent years, Pix2pixGAN has been put forward [24], [33], which has an excellent performance in image-to-image style translation. Compared with the three methods mentioned above, Pix2pixGAN makes the segmented image more accurate in detail by introducing a discriminator with a specific structure called PatchGAN. The discriminator can constrain the generator on small patches rather than on the whole image, making it pay more attention to details.

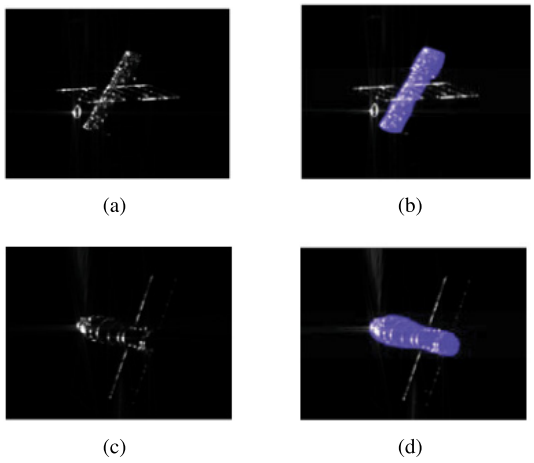


Fig. 3. ISAR image segmentation examples of satellite targets (the first row shows the KH satellite images; the second row shows the TG satellite images). (a) and (c) Original ISAR images. (b) and (d) Labeled ISAR images.

Inspired by the powerful capability of Pix2pixGAN, the ISAR image segmentation problem can be treated as an image-to-image style translation problem, as shown in Fig. 3. Fig. 3 shows two image sets: the original ISAR images and the labeled ISAR images. Pix2pixGAN can learn a loss function appropriate to train a mapping from the original ISAR image set to the labeled ISAR image set. Then, the output ISAR images, i.e., the images with a labeled satellite's capsule, can be used to segment the satellite's capsule from the initial ISAR images.

The basic idea of the proposed segmentation method is to learn the mapping function between two domains, X and Y, where the former implies the original ISAR images of the satellite, and the latter implies the labeled ISAR images. Given training samples, $\{x_i\}_{i=1}^N$ and $\{y_i\}_{i=1}^M$, where $x_i \in X$ and $y_i \in Y$, as shown in Fig. 4. The generator G is trained to produce outputs that cannot be distinguished from "real" images (which means the labeled ISAR images) by an adversarially trained discriminator, D, which is trained to do as well as possible at detecting the generator's "fakes." As shown in Fig. 4, the initial image of the satellite x and the noise vector z are inputted into the generator. The output image $G(x, z)$, together with the initial image x, is one paired image, and the labeled image y, together with the initial image x, is another paired image. These two sets of paired images are inputted into the discriminator to tell whether the generated image $G(x, z)$ can be distinguished from the real labeled image y. With the processing of training, the generator can generate images similar to the labeled images so that the discriminator cannot distinguish them.

The loss function of Pix2pixGAN consists of two parts: the traditional adversarial loss function and the L_1 loss function. The traditional adversarial is expressed as

$$\mathcal{L}_{\text{cGAN}}(G, D) = \mathbb{E}_{x_p, y_p}[\log D(x_p, y_p)] + \mathbb{E}_{x_p, z}[\log(1 - D(x_p, G(x_p, z))] \quad (5)$$

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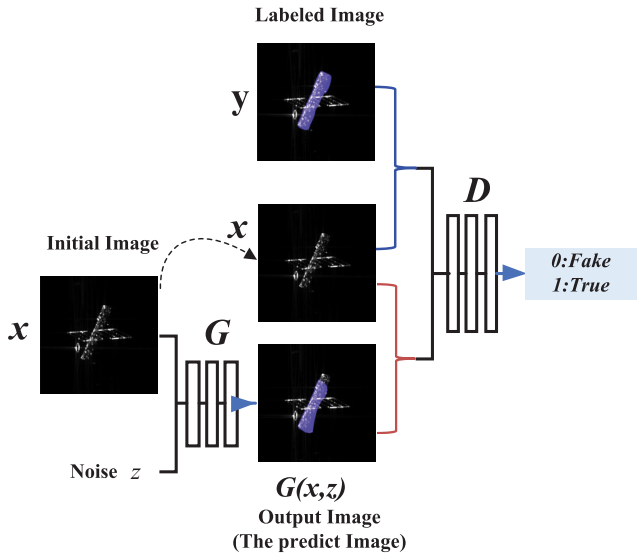


Fig. 4. Training process of Pix2pixGAN.

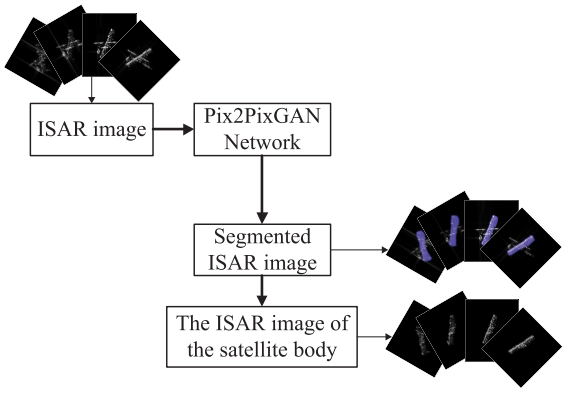


Fig. 5. Flowchart of the proposed ISAR image segmentation method.

where x_p and y_p are the patches of the initial images and the labeled images, $p = 1, 2, \ldots, P$, and P is the total number of patches. G tries to minimize this objective against an adversarial D that tries to maximize it, i.e., $G^* = \arg\min_G \max_D \mathcal{L}_{\text{cGAN}}(G, D)$.

The L_1 loss function is expressed as

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y,z}[\|y - G(x, z)\|_1]. \tag{6}$$

Thus, the final objective of Pix2pixGAN is

$$G^* = \arg\min_G \max_D \mathcal{L}_{\text{cGAN}}(G, D) + \lambda \mathcal{L}_{L1}(G). \tag{7}$$

After getting a well-trained Pix2pixGAN network based on the existing ISAR images, the well-trained network can be treated as an image segmentation tool. When a new ISAR image arrives, the corresponding colored image can be obtained by inputting the new ISAR image into the network. The satellite's capsules can be obtained by finding the colored pixels in the initial ISAR images, as shown in Fig. 5.

It should be noted that, although the proposed segmentation method performs well, the segmentation does not achieve pixel-level accuracy. The output of the Pix2pixGAN network only marks the region where the satellite capsule is located, which does not mean that the marked region exactly matches the satellite capsule. If we extract the projection vector $\mathbf{N}_{\text{body}}^{\text{project}}$ directly based on the marked area, it will lead to a large error in the extracted features. Therefore, in the following, we propose the OMP-based scatterers extraction and PCA-based feature extraction method.

B. Scatterers Extraction Using OMP Algorithm

After the ISAR image segmentation process has been done, we can get an image of the satellite's capsule part, which can be used to extract the scatterers' locations on the satellite's capsule in an ISAR image plane. The scatterers' locations can be used to estimate the satellite capsule's 2-D attitude in Section III-C. As discussed in [10] for compressive sensing (CS)-based ISAR imaging, the signal can be approximated by the echoes from a few individual strong scattering centers, and those from the weak ones are regarded as noise. Considering this, we can extract scattering centers based on CS, commonly used approaches are OMP. Li and Du [34] have introduced the method to extract scattering centers based on OMP; here, we give a brief introduction to this approach. First, an object's total backscattering response can be approximated as a sum of responses from individual scattering centers. The backscattered field from a target can be expressed as

$$\mathbf{E}(f, \phi; \hat{\theta}) = \sum_{i=1}^{N} \mathbf{E}_i(f, \phi; \theta_i) \tag{8}$$

where

$$\mathbf{E}_i(f, \phi; \theta_i) = A_i \cdot \exp\left[-j\frac{4\pi f}{c}(x_i \cos\phi + y_i \sin\phi)\right]. \tag{9}$$

In (9), $\mathbf{E}_i(f, \phi; \theta_i)$ is the response signal from the ith scattering center, and $\theta_i = [A_i, x_i, y_i]^T$ is the parameter vector of scattering center i, where A_i and (x_i, y_i) represent the amplitude and location, respectively. Then, c is the speed of light, N is the number of scattering centers, and the parameter matrix of these N individual scattering centers is $\theta = [\theta_1, \theta_2, \ldots, \theta_N]^T$. The sparse representation of signal from N individual scattering centers can be expressed as

$$\mathbf{s} = \mathbf{D}(\theta)\sigma + \eta \tag{10}$$

where $\mathbf{s}$ is the vectorization of measurements $\mathbf{E}(f, \phi; \hat{\theta})$, σ is the coefficient vector, $\mathbf{D}(\theta)$ is a parameterized redundant dictionary, each column of dictionary $\mathbf{D}(\theta)$ is the vectorization of measurements corresponding to one element in parameter set θ, and η represents the additive noise. According to sparse representation theory, scattering centers can be extracted by solving the following sparse optimization problem:

$$\hat{\sigma} = \arg\min_{\sigma} \|\sigma\|_0, \quad \text{s.t. } \|\mathbf{s} - \mathbf{D}(\theta)\sigma\|_2^2 \le \varepsilon \tag{11}$$

where $\|\cdot\|_0$ denotes the l_0 norm, $\|\sigma\|_0$ represents the number of nonzero entries in vector σ, $\hat{\sigma}$ is the estimated coefficient, and ε is the noise level. Considering that a small number of scattering centers contribute most of the energy in radar echo, radar echo has a strong sparsity in parameter space, which

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implies that most elements in σ are zeros. OMP is an iterative greedy algorithm that selects the column of $\mathbf{D}(\theta)$, which is the most correlated with the current residual at each step. In each iteration, one parameter vector θ_i corresponding to a scattering center can be obtained. After N iterations, all the scattering centers are acquired in parameter set $\theta = [\theta_1, \theta_2, \ldots, \theta_N]^T$, and the number of iterations can be set based on scattering center number N or the noise energy. The detailed procedure of the OMP algorithm for obtaining scattering centers is given as follows.

1) Step 1: **Inputs**: The measurements $\mathbf{s}$, the estimated noise level ε, or the iteration number N.
2) Step 2: **Initialization**: The initial residual signal $\mathbf{r}_0 = \mathbf{s}$, and the iteration counter $k = 1$, and the parameters set of the scattering centers $\hat{\theta} = \emptyset$.
3) Step 3: Build the redundant parameterized dictionary $\mathbf{D}(\theta)$

$$\mathbf{D}(\theta) = [\mathbf{d}_1, \ldots, \mathbf{d}_i, \ldots, \mathbf{d}_M] \tag{12}$$

where $i = 1, 2, \ldots, M$, and

$$\mathbf{d}_i = \frac{vec(\mathbf{E}_i(f, \phi; \theta_i))}{\|vec(\mathbf{E}_i(f, \phi; \theta_i))\|_2}, \quad \theta_i \in \theta \tag{13}$$

In (12) and (13), M is the total number of dictionary atoms in $\mathbf{D}(\theta)$, vec$(\cdot)$ represents the vectorization operation, and $\|\text{vec}(\mathbf{E}_i(f, \phi; \theta_i))\|_2$ is the normalization of $vec(\mathbf{E}_i(f, \phi; \theta_i))$.
4) Step 4: If $\|r_0\|_2^2 > \varepsilon$ or iteration counter $k < N$, go to Step 5.
5) Step 5: Calculate Correlation: $\mathbf{C}(\theta) = \mathbf{D}^H(\theta) \times \mathbf{r}_0$, where $(\cdot)^H$ denotes the conjugate transpose.
6) Step 6: Obtain the parameter set of scattering center: $\hat{\theta}_k = \arg\max_\theta \mathbf{C}(\theta)$, $\hat{\theta} = \hat{\theta} \cup \hat{\theta}_k$.
7) Step 7: Estimate the amplitude σ_k of scattering center k using the least-squares method (LSM).
8) Step 8: Update the residual signal $\mathbf{r}_0$: $\mathbf{r}_0 = \mathbf{s} - \mathbf{D}(\hat{\theta})\sigma_k$, and iteration counters $k = k + 1$; go back to Step 4.

After the extraction process using OMP, we can get the parameter set $\hat{\theta} = [\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_N]^T$ corresponding to each scattering center, which contains the location parameters of each point (x_i, y_i). These location parameters will be used to estimate the 2-D orientation of the satellite's capsule in Section III-C.

C. Feature Estimation Based on PCA

In Section III-B, we have got the scatterers locations of the satellite's capsule in the ISAR image, i.e., (x_i, y_i), $i = 1, 2, \ldots, N$. These scatterers' coordinates can be used to estimate the 2-D orientation of the satellite's capsule in the ISAR image via PCA. PCA transforms a dataset into one in a new coordinate system where the largest variance corresponds to the first coordinate [18]. Using this property of PCA, the first principal component has the largest variance, which accounts for as much of the datasets' variability. We focus on the fact that there exists a unique direction with the largest variance corresponding to the satellite's capsule orientation. As we have mentioned in Section II, the proposed method applies

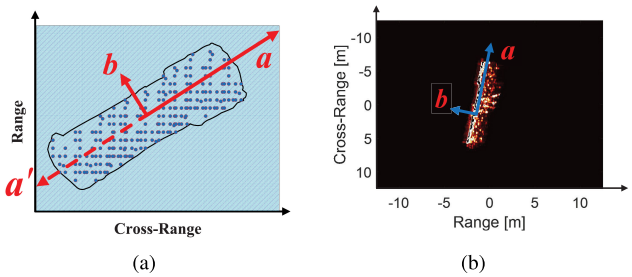


Fig. 6. Geometry of scattering centers distribution and its principal components. (a) Satellite's capsule and its principal component. (b) Simulated satellite's capsule.

to satellites with symmetrical capsules, such as cylinders and cuboids. These special structures ensure that the satellite capsule's projection on the ISAR image is also symmetrical, as shown in Fig. 6(a). The direction of eigenvector $\mathbf{n_1}$ corresponding to the largest eigenvalue represents the major axis of the dataset that includes distributed scattering centers. Eigenvector $\mathbf{n_1}$ is what we needed, which refers to the 2-D attitude of the satellite's capsule $\mathbf{N}_{\text{body}}^{\text{project}}$ in (3) Section II.

Let

$$\mathbf{W} = \begin{bmatrix} x_1 & x_2 & \cdots & x_i & \cdots & x_N \\ y_1 & y_2 & \cdots & y_i & \cdots & y_N \end{bmatrix}$$

denote the location matrix of the scattering centers extracted from a segmented ISAR image. In general, the first step is to shift each row of $\mathbf{W}$ to make it has a zero mean, which is realized by the following equation:

$$\begin{aligned} \bar{\mathbf{W}} &= \mathbf{W} - \text{mean}(\mathbf{W}) \\ &= \begin{bmatrix} x_1 - \bar{x} & x_2 - \bar{x} & \cdots & x_i - \bar{x} & \cdots & x_N - \bar{x} \\ y_1 - \bar{y} & y_2 - \bar{y} & \cdots & y_i - \bar{y} & \cdots & y_N - \bar{y} \end{bmatrix} \end{aligned} \tag{14}$$

where $\bar{x}$ is the mean value of $\{x_i\}_{i=1,2,\ldots,N}$ and $\bar{y}$ is the mean value of $\{y_i\}_{i=1,2,\ldots,N}$. The covariance matrix can be obtained by $\mathbf{C} = \bar{\mathbf{W}} \cdot \bar{\mathbf{W}}^T$. Then, the covariance matrix undergoes eigenvalue decomposition, i.e., $\mathbf{C} = \mathbf{E}\Lambda\mathbf{E}^T$ to bring out the principle component, where Λ is a diagonal matrix comprised by the eigenvalues and $\mathbf{E}$ is a matrix containing the corresponding eigenvectors. The first principal component is the eigenvector corresponding to the largest eigenvalue, which implies the orientation of the projected satellite's capsule in a 2-D imaging plane, i.e., shown as $\mathbf{a}$ in Fig. 6(b). Here, we use $\hat{\mathbf{N}}_{\text{body}}^{\text{project}}$ to represent the extracted unit 2-D orientation vector of the satellite's capsule, as we do in (3), and it can be expressed as $\hat{\mathbf{N}}_{\text{body}}^{\text{project}} = (\mathbf{a}/\|\mathbf{a}\|)$. Considering that the obtained orientation vectors $\hat{\mathbf{N}}_{\text{body}}^{\text{project}}$ may be in the opposite directions for differents ISAR images, shown as $\mathbf{a}$ and $\mathbf{a}'$ in Fig. 6(b), some rules need to be defined. As shown in Fig. 6(a), if we define the orientation of satellite capsule's head (the same direction as $\mathbf{a}$) as positive, then the orientation vector can be expressed as $\hat{\mathbf{N}}_{\text{body}}^{\text{project}} = (\mathbf{a}/\|\mathbf{a}\|)$; else, if we define the orientation of satellite capsule's tail (the same direction as $\mathbf{a}'$) as positive, $\hat{\mathbf{N}}_{\text{body}}^{\text{project}} = (\mathbf{a}'/\|\mathbf{a}'\|)$. Then, the projected length of the satellite's capsule can be estimated by projecting all the extracted scatterers onto the vector $\mathbf{a}$, as shown in Fig. 7. The

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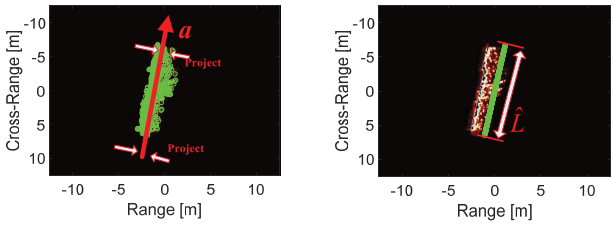


Fig. 7. Extracted length of the satellite's capsule. (a) Satellite's capsule and extracted scatterers' centers. (b) Projected length of the satellite's capsule.

projected length $\hat{L}$ can be estimated as follows:

$$\hat{L} = \max\left(\mathbf{W}^T \cdot \mathbf{a}\right) - \min\left(\mathbf{W}^T \cdot \mathbf{a}\right). \tag{17}$$

As we described in (4), the estimated vector of the satellite's capsule can be expressed as

$$\hat{\mathbf{n}} = \hat{L} \cdot \hat{\mathbf{N}}_{\text{body}}^{\text{project}}. \tag{18}$$

So far, the orientation and length of the projected satellite's capsule have been extracted from each frame of the ISAR image. The estimated orientation $\hat{\mathbf{N}}_{\text{body}}^{\text{project}}$ and length $\hat{L}$ will be used to find the optimal solution of the absolute attitude and length of the satellite's capsule in Section III-D. It should be emphasized that, since the extraction of the scatterers of the satellite's capsule is mainly affected by the image resolution, this will directly affect the features' extraction accuracy of the projection orientation $\hat{\mathbf{N}}_{\text{body}}^{\text{project}}$ and the projection length $\hat{L}$ of the satellite's capsule. In the low-resolution case, the satellite's capsule occupies fewer pixel points The extracted scatterers cannot provide a better description of the satellite's capsule structure, which will increase the $\hat{\mathbf{N}}_{\text{body}}^{\text{project}}$ and $\hat{L}$ extraction errors and lead to the poor performance of the proposed algorithm in estimating the satellite attitude and size. In general, most of the existing radars for space target observation emit large bandwidth signals with high resolution, so the accuracy of $\hat{\mathbf{N}}_{\text{body}}^{\text{project}}$ and $\hat{L}$ feature extraction is generally satisfied in practical situations.

D. Parameters' Estimation by an Iterative Optimization

Here, we define the distance between vector $\mathbf{n}(\theta, \varphi; L)$ and vector $\hat{\mathbf{n}}$ as an optimization function, as follows:

$$f(\theta, \varphi; L) = \sum_{i=1}^{N} \|\mathbf{n}_i(\theta, \varphi; L) - \hat{\mathbf{n}}_i\| \tag{19}$$

where $(\theta, \varphi; L)$ are the parameters that need to be estimated, i is the frame index of ISAR images, and $i = 1, 2, \ldots, N$, $\|\cdot\|$ represents a vector modulo operation. The estimation of $(\theta, \varphi; L)$ is obtained by minimizing the fitness function, i.e., $f(\theta, \varphi; L)$, and the optimization function can be expressed as follows:

$$\left\langle \theta^*, \varphi^*, L^* \right\rangle = \arg\min_{\theta, \varphi, L} f(\theta, \varphi; L). \tag{20}$$

To solve this optimization, many standard algorithms are available, such as PSO [18], [35] and genetic algorithms. However, these nonparametric methods always need great computation time. In this article, to increase the computation efficiency, a gradient-based iterative approach is adopted.

First, we need to get the partial derivatives of each parameter. $f(\theta, \varphi; L)$ can be reexpressed as

$$f(\theta, \varphi; L) = \sum_{i=1}^{N} \sqrt{\left[\mathbf{n}_i(1) - \hat{\mathbf{n}}_i(1)\right]^2 + \left[\mathbf{n}_i(2) - \hat{\mathbf{n}}_i(2)\right]^2} \tag{21}$$

where $\mathbf{n}_i(1)$ and $\mathbf{n}_i(2)$ represent the first and second elements of vector $\mathbf{n}_i(\theta, \varphi; L)$; $\hat{\mathbf{n}}_i(1)$ and $\hat{\mathbf{n}}_i(2)$ represent the first and second elements of vector $\hat{\mathbf{n}}_i$. According to (4), $\mathbf{n}_i(1)$ and $\mathbf{n}_i(2)$ can be expressed as

$$\begin{cases} \mathbf{n}_i(1) = \mathbf{E}_1^T L \cdot \mathbf{P}_i \cdot \mathbf{N}_{\text{body}} \\ \mathbf{n}_i(2) = \mathbf{E}_2^T L \cdot \mathbf{P}_i \cdot \mathbf{N}_{\text{body}} \end{cases} \tag{22}$$

where $\mathbf{E}_1 = [1\ 0]^T$ and $\mathbf{E}_2 = [0\ 1]^T$. Then, the partial derivation of θ can be expressed as (15), shown at the bottom of the page.

In Section II, $\mathbf{N}_{\text{body}} = [\cos\varphi \sin\theta, \cos\varphi \cos\theta, \sin\varphi]^T$, so $(\partial \mathbf{N}_{\text{body}}/\partial\theta)$ can be expressed as follows:

$$\frac{\partial \mathbf{N}_{\text{body}}}{\partial \theta} = [\cos\varphi \cos\theta, -\cos\varphi \sin\theta, 0]^T. \tag{23}$$

Similar to the derivation of θ, the partial derivative of φ and L can be obtained. From (15) and (23), we can see that the difference between the derivation of θ and φ is the $(\partial \mathbf{N}_{\text{body}}/\partial)$ term, as, to φ, the expression of this term is expressed as follows:

$$\frac{\partial \mathbf{N}_{\text{body}}}{\partial \varphi} = [-\sin\varphi \sin\theta, -\sin\varphi \cos\theta, \cos\varphi]^T. \tag{24}$$

As to L, the derivation of L can be expressed as (16), shown at the bottom of the page.

$$\begin{aligned} \frac{\partial f(\theta, \varphi; L)}{\partial \theta} &= \sum_{i=1}^{N} \frac{\left[\mathbf{n}_i(1) - \hat{\mathbf{n}}_i(1)\right] \cdot \frac{\partial \mathbf{n}_i(1)}{\partial \theta} + \left[\mathbf{n}_i(2) - \hat{\mathbf{n}}_i(2)\right] \cdot \frac{\partial \mathbf{n}_i(2)}{\partial \theta}}{\sqrt{\left[\mathbf{n}_i(1) - \hat{\mathbf{n}}_i(1)\right]^2 + \left[\mathbf{n}_i(2) - \hat{\mathbf{n}}_i(2)\right]^2}} \\ &= \sum_{i=1}^{N} \frac{\left[\mathbf{n}_i(1) - \hat{\mathbf{n}}_i(1)\right] \cdot \left(\mathbf{E}_1^T L \cdot \mathbf{P}_i \cdot \frac{\partial \mathbf{N}_{\text{body}}}{\partial \theta}\right) + \left[\mathbf{n}_i(2) - \hat{\mathbf{n}}_i(2)\right] \cdot \left(\mathbf{E}_2^T L \cdot \mathbf{P}_i \cdot \frac{\partial \mathbf{N}_{\text{body}}}{\partial \theta}\right)}{\sqrt{\left[\mathbf{n}_i(1) - \hat{\mathbf{n}}_i(1)\right]^2 + \left[\mathbf{n}_i(2) - \hat{\mathbf{n}}_i(2)\right]^2}} \end{aligned} \tag{15}$$

$$\frac{\partial f(\theta, \varphi; L)}{\partial L} = \sum_{i=1}^{N} \frac{\left[\mathbf{n}_i(1) - \hat{\mathbf{n}}_i(1)\right] \cdot \left(\mathbf{E}_1^T \mathbf{P}_i \cdot \mathbf{N}_{\text{body}}\right) + \left[\mathbf{n}_i(2) - \hat{\mathbf{n}}_i(2)\right] \cdot \left(\mathbf{E}_2^T \mathbf{P}_i \mathbf{N}_{\text{body}}\right)}{\sqrt{\left[\mathbf{n}_i(1) - \hat{\mathbf{n}}_i(1)\right]^2 + \left[\mathbf{n}_i(2) - \hat{\mathbf{n}}_i(2)\right]^2}} \tag{16}$$

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TABLE I
RADAR PARAMETERS AND LOCATION

Radar Location	(116.4E,39.9N,0m)	
Radar Parameters	Center Frequency	16.67GHz
	Pulse Repetition Frequency	25Hz
	Sample Frequency	2.4GHz
	Bandwidth	2GHz

With the partial derivative expressions in (15), (23), and (24), the gradient of loss function $f(\theta, \varphi; L)$ for (θ, φ, L) is

$$\Delta \mathbf{f} = \left[\frac{\partial f(\theta, \varphi; L)}{\partial \theta}, \frac{\partial f(\theta, \varphi; L)}{\partial \varphi}, \frac{\partial f(\theta, \varphi; L)}{\partial L} \right]^T. \quad (25)$$

In general, the second-order derivative is needed to build the Hessian matrix if the gradient descent algorithms are applied here. Considering the computation complicity of second-order derivatives, the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm [36], [37] is adopted in this article to search the optimal solution. Let $\mathbf{x}_0 = [0, 0, 0]^T$ be the initial parameter vector and $\mathbf{x}_k$ be the parameter vector of the kth iteration. In BFGS, the searching direction in each iteration can be calculated as follows:

$$\mathbf{d}_k = \mathbf{B}_{k-1} \Delta \mathbf{f}_k. \quad (26)$$

Then, the $k+1$th parameter vector can be expressed as

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \lambda_k \mathbf{d}_k \quad (27)$$

where λ_k is the searching step that can be estimated by (28) via some 1-D inexact searching methods, such as the Golden search and the Armijo–Goldstein principle

$$\lambda_k = \arg\min_{\lambda_k} (f(\mathbf{x}_k + \lambda_k \mathbf{d}_k)). \quad (28)$$

In (22), $\mathbf{B}_{k-1}$ is updated as follows:

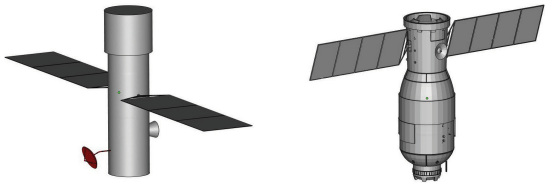
$$\mathbf{B}_k = \mathbf{B}_{k-1} + \frac{\mathbf{y}_k \cdot \mathbf{y}_k^T}{\mathbf{y}_k \cdot \mathbf{s}_k^T} - \frac{\mathbf{B}_{k-1} \mathbf{s}_k \mathbf{s}_k^T \mathbf{B}_{k-1}}{\mathbf{s}_k^T \mathbf{B}_{k-1} \mathbf{s}_k} \quad (29)$$

where $\mathbf{s}_k = \lambda_k \mathbf{d}_k$ and $\mathbf{y}_k = \Delta \mathbf{f}_{k+1} - \Delta \mathbf{f}_k$. The iteration can be terminated when $\|\Delta \mathbf{f}_k\| < \varepsilon$ or $|f(\mathbf{x}_{k+1}) - f(\mathbf{x}_k)| < \varepsilon$. After we get the optimal solution $\mathbf{x}_k$, the absolute attitude and the length of the satellite's capsule can be obtained.

IV. SIMULATIONS RESULTS

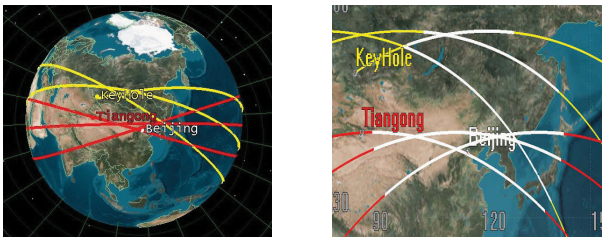
A. Introduction of Experimental Data

Before starting the experiment, we provide a brief description of the simulation data used in the experiment. Due to the lack of measured data from satellites, the experiment was analyzed using simulated ISAR data. The radar parameters of a Ku-band ISAR system are listed in Table I. Fig. 8 gives the computer-aided-design (CAD) models of those two satellite targets, the KeyHole (KH) satellite and the Tiangong (TG) satellite, which is used to generate data for the following experiments.



(a) (b)

Fig. 8. CAD models of two satellite targets. (a) CAD model of the KH satellite. (b) CAD model of the TG satellite.



(a) (b)

Fig. 9. Orbits of TG and KH satellites simulated by STK. (a) Trajectory of these satellites that circle the Earth three times. (b) Trajectory segments observable by radar (white).

TABLE II
DURATION OF RADAR OBSERVATIONS AND THE NUMBER OF ISAR IMAGES ACQUIRED

	Observation Time [s]	Image Size	Total Frame	Image Resolution [m]
Lap1	708.22	512*512	51	0.075*0.075
Lap2	679.78		49	
Lap3	473.24		42	

The satellite tool kits (STKs) are used to simulate the real orbital motion of these two satellites. In the experiment, STK simulates the radar LOS of a satellite continuously circling the Earth three times. Fig. 9(a) gives the ground track of these two satellites, and Fig. 9(b) shows the trajectory segments that the radar can observe. STK can simulate the real orbit movement of these two satellites to get the radar LOS during observation time [11]. After getting the radar LOS, the electromagnetic simulation software FEKO is employed to generate the radar echoes of these two satellite targets, and the SNR of the echo is set to 20 dB.

Table II shows the duration of radar observations for each lap and the frames of ISAR images generated from these observation data. The traditional range-Doppler (RD) algorithm generates all the ISAR images, and the Pix2pixGAN network is trained by those images obtained via the first two laps of observation data (lap 1 and lap 2). The images obtained via the last lap (lap 3) data are used to estimate absolute attitude and length parameters.

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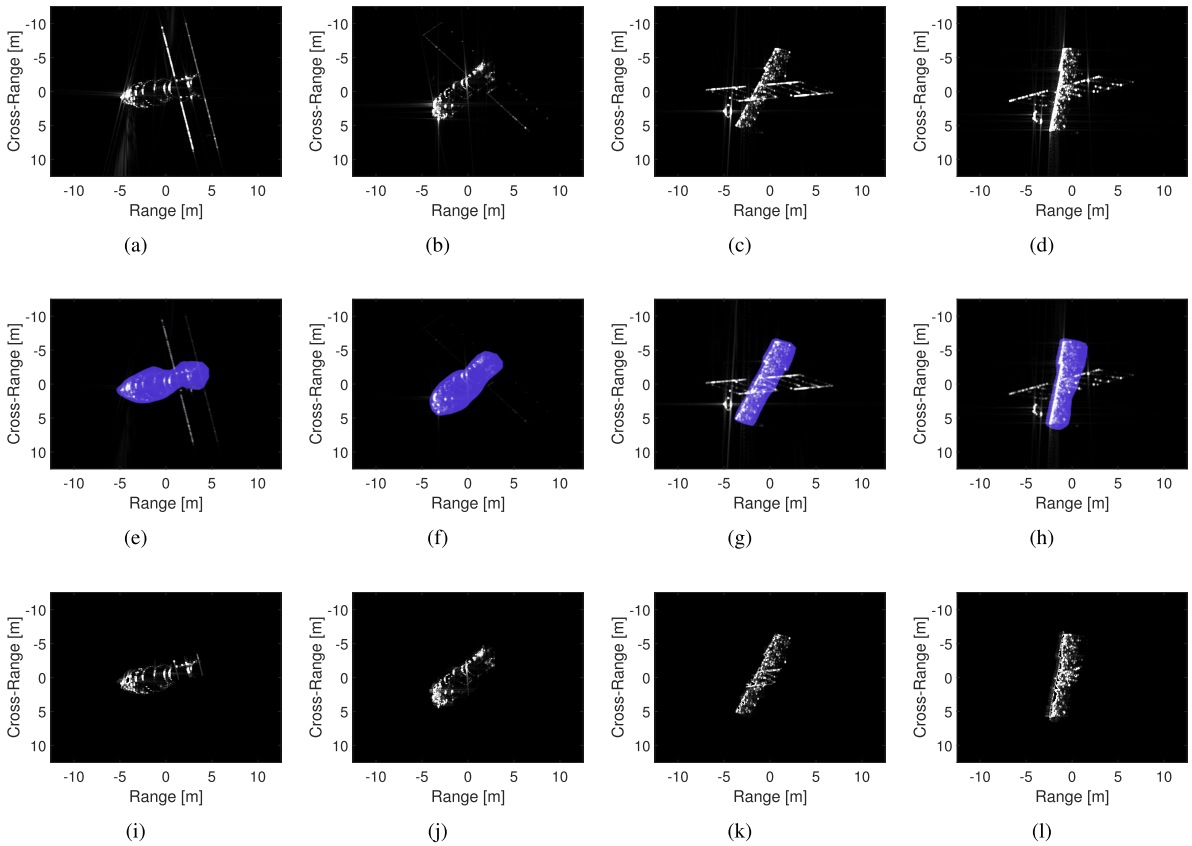


Fig. 11. Some examples of the segmentation outputs. (a)–(d) Initial ISAR images. (e)–(h) Output of the Pix2pixGAN network (labeled images). (i)–(l) Segmented satellite's capsule. (a), (e), and (i) 18th frame of TG. (b), (f), and (j) 42nd frame of TG. (c), (g), and (k) 25th frame of KH. (d), (h), and (l) 35th frame of KH.

B. Performance of the Proposed ISAR Image Segmentation Method

After acquiring the ISAR images of lap 1 and lap 2, these images need manual processing before training the Pix2pixGAN network. Considering the lack of training data, some augmentation operations should be done beforehand. In this article, each image undergoes the procedure of translation and flipping. To expand the training images to 36 times (3600 frames of each target) of the initial images (100 frames), after increasing the number of images, the initial ISAR images and the labeled images are input into the Pix2pixGAN network to train the mapping from initial images to labeled images. It should be noted that, when training the Pix2pixGAN network, all the target (here, KH and TG) images are input to the network together for training.

To demonstrate the advantages of the proposed ISAR image segmentation algorithm, different segmentation algorithms are applied herein. The segmentation accuracy of these segmentation algorithms is also compared. Table III gives a comparison between the segmentation accuracy of the proposed segmentation algorithm and other segmentation algorithms. The criterion that we use here for segmentation accuracy is called mean intersection over union (MIoU) [38], which is the standard metric for segmentation purposes. It computes a ratio between the intersection and the union of two sets: the ground truth and our predicted segmentation. The higher the value of MIoU, the higher the segmentation accuracy. Fig. 10 shows the MIoU diagram of ISAR image segmentation. The MIoU is computed as follows:

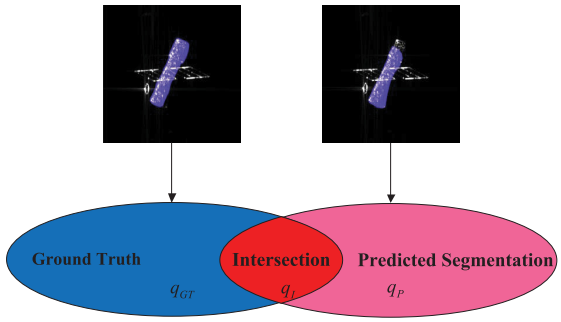


Fig. 10. MIoU interpretation of ISAR image segmentation. The left ISAR image is the ground truth, and the right ISAR image is the segmented image predicted by segmentation algorithms.

$$\text{MIoU} = \frac{1}{k}\sum_{i=1}^{k}\frac{q_I}{q_{\text{GT}} + q_P - q_I} \tag{30}$$

where q_I represents the number of intersection pixels, q_{GT} represents the number of pixels that belong to ground truth, q_P represents the number of pixels belongs to predicted

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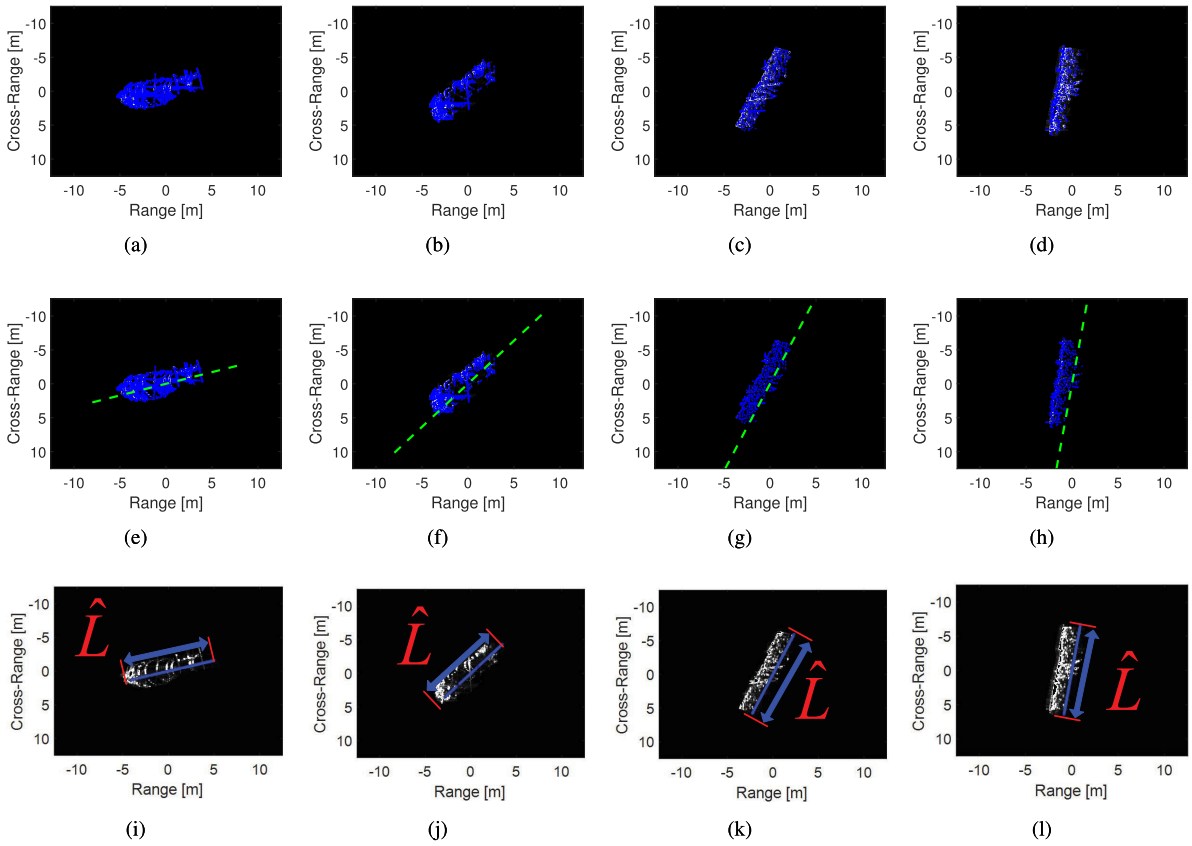


Fig. 12. Feature extraction from the segmented satellite's capsules. (a)–(d) Obtained scatterers of the satellite's capsules. (e)–(h) 2-D orientation of the satellite's capsules. (i)–(l) Projected capsule lengths of the satellite.

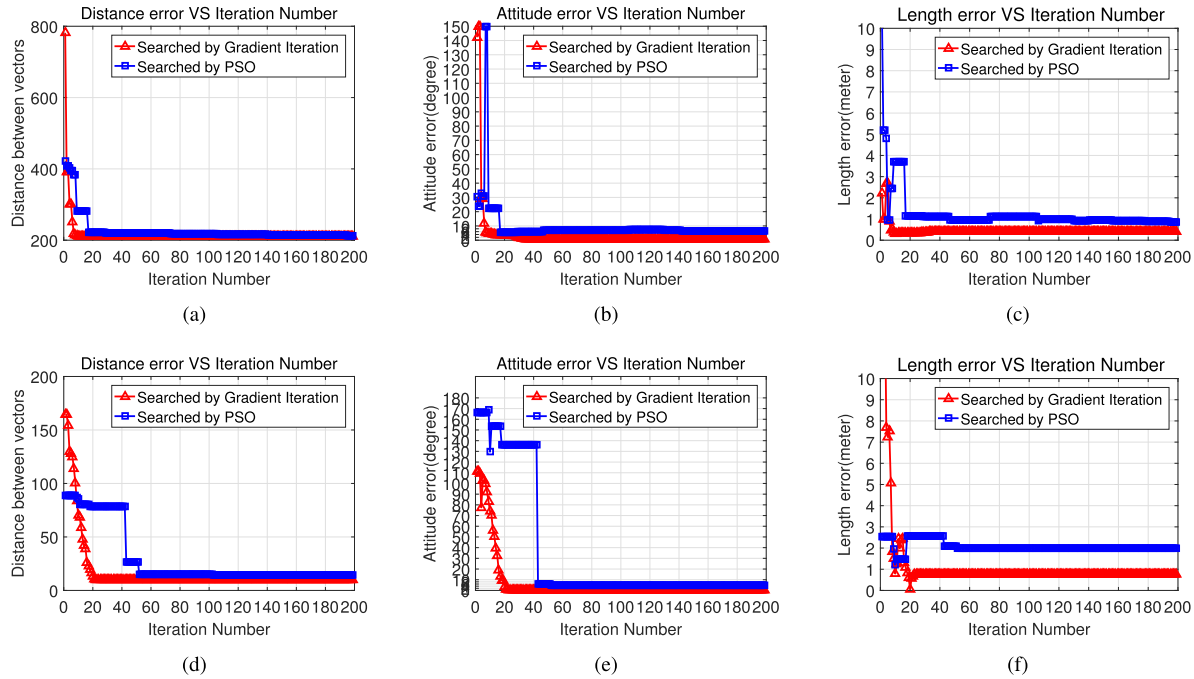


Fig. 13. Optimization functions and the variation of size and attitude estimation errors with the number of iterations. (a) and (d) Variation curves of optimization functions with the number of iterations for KH and TG satellites. (b) and (e) Variation curves of attitude estimation errors with the number of iterations for KH and TG satellites. (c) and (f) Variation curves of size errors with the number of iterations for KH and TG satellites.

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TABLE III
COMPARISON OF MIOU RESULTS OF DIFFERENT METHODS

Targets	Methods			
	FCN	SegNet	DeepLabv3+	Proposed Method
KeyHole(KH)	76.55	78.53	78.26	80.01
Tiangong (TG)	75.03	77.08	75.12	77.94
Average	75.79	77.81	76.69	78.98

segmentation, and k is the total number of ISAR images. From Table III, it can be seen that the proposed segmentation algorithm has a higher value of MIoU compared to other segmentation algorithms, which means that the proposed segmentation method has a higher segmentation accuracy. As we have mentioned in Section III-A, the Pix2pixGAN makes ISAR image segmentation more accurate by introducing a specially structured discriminator called PatchGAN that allows the generator to pay more attention to detail during the segmentation process. That is why the proposed segmentation method has a higher MIoU value than other segmentation methods.

C. Performance Investigation

1) Performance of the Propose Algorithm: In this section, the proposed method's performance will be evaluated with simulated data of TG and KH satellites. As we have mentioned in Section IV-A, ISAR images acquired on the third lap will be used to estimate the absolute attitude and size of the satellite. First, all the images (42 frames) are input into the well-trained Pix2pixGAN network for labeling the satellite's capsule, as shown in Fig. 11. In Fig. 11, the first row is the initial ISAR images of TG, and the second row is the labeled images that output from the Pix2pixGAN network. As one can note, the capsule area in the image has perfectly colored with blue, and by selecting the colored pixels in the corresponding ISAR images, we can get the capsule parts of the satellite, as shown in the third row.

After acquiring the capsule part of the satellite, the OMP algorithm is employed to extract the scatterers' position on the satellite's capsules. It should be noted that the iteration number should be well determined to ensure that the scatterers on the satellite's capsules are all extracted. The obtained scatterers of the satellite's capsules, i.e., (x_i, y_i), are shown in the first row in Fig. 12. Based on the obtained locations of the scatterers, we can get the location matrix $\mathbf{W}$. Then, the location matrix $\mathbf{W}$ will undergo PCA operation, which has been mentioned in Section III-C, the 2-D orientation of the satellite's capsules can be obtained. The estimated 2-D orientations are shown in the second row in Fig. 12. As we can see, the estimated orientation vectors are consistent with the orientations of the satellite's capsules. Then, the obtained scatterers are projected onto the first principal component using (17). The projected scatterers can be seen in the third row in Fig. 12. The projected length of the satellite's capsule can be estimated by (17).

After the feature estimation process, we can get a dataset $\{\hat{\mathbf{N}}_{\text{body},i}^{\text{project}}\}$, the capsules 2-D orientation unit vectors, and a dataset $\{\hat{L}_i\}$, the capsule lengths, where i represents the frame index, $i = 1, 2, \ldots, F$, and F is the total frame number. As shown in (3), these two datasets can be used to represent the projected capsule vector, which are $\{\mathbf{n}_i\} = \hat{L}_i \cdot \hat{\mathbf{N}}_{\text{body},i}^{\text{project}}$. As has been introduced in Section III-D, the projected capsule vectors will be input into the fitness function to estimate the satellite absolute attitude and size iteratively. For comparison, we also present the estimation results using the PSO algorithm. In the estimation process, the number of iterations for both algorithms is set to 200, fully satisfying that the fitness function converges to the optimal solution. The estimated attitude vectors and lengths of the satellite's capsule are listed in Table IV; we also give the estimated errors in absolute attitude and size, calculated as (31), where $\mathbf{N}_{\text{body}}^T$ denotes the true unit vector of the satellite's capsule, i.e., the real absolute attitude of the satellite. $\hat{\mathbf{N}}_{\text{body}}$ is the estimated attitude of the satellite's capsule. From Table IV, we can see that, for the two satellite targets with different postures, the proposed algorithm can estimate the capsule attitudes and lengths accurately to a certain degree. The attitude estimation error for both satellites is less than 0.5°, and the length errors are all below one meter. However, compared with the proposed algorithm, the PSO algorithm performs slightly worse. From Table IV, we can see that the estimation errors of PSO are higher than that of the proposed algorithm, whether they are attitudes or lengths, which shows the accuracy of the proposed algorithm

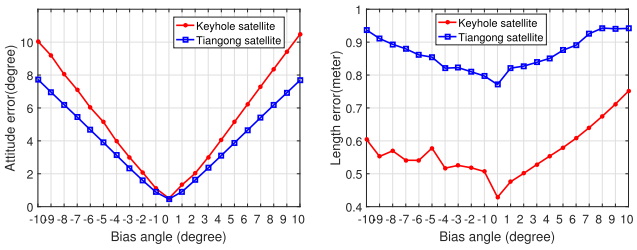
$$\begin{cases} \text{error}_{\text{pose}} = \arccos\left(\mathbf{N}_{\text{body}}^T \cdot \hat{\mathbf{N}}_{\text{body}}\right) \\ \text{error}_{\text{length}} = \left|L - \hat{L}\right|. \end{cases} \tag{31}$$

We also give the value of the fitness function and the curve of the estimation error with the number of iterations for both satellites, which are shown in Fig. 13. Fig. 13(a) and (d) shows the variations in distance errors (the fitness function values) between the true and estimated attitude of the satellite's capsule; Fig. 13(b) and (e) shows the variation in attitude errors; and Fig. 13(c) and (f) shows the length error variation. From these error variations, it can be seen that the proposed algorithm converges faster and estimates more accurately than the PSO algorithm. As one can see that, both algorithms are iterative algorithms. For the PSO algorithm, the amount of computation per iteration is related to the number of particles set to 6 in the experiment. Each particle denotes one calculation of (19) ($3N$ multiplications). Thus, the computation burden of each iteration for PSO is about $O(N)$. For the proposed searching method, each iteration denotes one calculation of (25) ($15N$ multiplications), (26)–(29) (69 multiplications), respectively; therefore, the computation burden of each iteration for the proposed method is also about $O(N)$. Thus, the computational complexity of these two searching methods is almost the same. Table V shows the computation time of these two methods obtained by running MATLAB codes on a personal computer with an Intel Core i5 3.30-GHz processer and 8-GB memory. As one can note, the operation times for a single iteration of the two algorithms are quite similar. However, compared with PSO, the proposed method

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TABLE IV
ATTITUDE AND SIZE ESTIMATIONS FOR THE PROPOSED ALGORITHM AND THE PSO ALGORITHM

Target		KeyHole (KH)		TianGong (TG)	
		Attitude	Length	Attitude	Length
True values		(0.500,0,0.866)	12 [m]	(0.163,-0.925,-0.342)	11.488 [m]
Proposed method		(0.499,0.005 ,0.867)	12.405 [m]	(0.164,-0.927,-0.337)	12.260 [m]
PSO		(0.519,0.112,0.848)	12.864 [m]	(0.149,0.897,0.417)	13.465 [m]
Errors	Proposed method	0.493°	0.405 [m]	0.465°	0.771 [m]
	PSO	6.620°	0.864 [m]	4.673°	1.977 [m]



(a) (b)

Fig. 14. Variation of attitude and length estimation errors with bias error of PCA feature extraction. (a) Variation of attitude estimation error with bias. (b) Variation of length estimation error with bias.

needs fewer iterations and computation time to achieve convergence. The reason is that, compared with the gradient-based search method, PSO is an exhaustive search method. Unlike the gradient-based search method, PSO does not search in the optimal direction during each iteration, resulting in slow convergence and imprecise search. However, the proposed method requires fewer iterations to reach the optimal solution, saving a lot of computation time.

2) Error Analysis: In this work, the accuracy of the capsule features extracted by PCA is the main factor affecting the algorithm's performance. The presence of the surface scatterers mechanism leads to bias between the extracted projected direction of the capsule and true projected direction, and such bias affects the subsequent absolute attitude and size estimation of the satellite. To investigate the sensitivity of the proposed algorithm against extraction bias error, we added a directional bias from $-10°$–$10°$ to the ground truth of the satellite capsule projected 2-D direction. The 3-D absolute attitude and length errors are estimated with the proposed method under different biases, and the results are shown in Fig. 14. It can be seen from the figure that, as the bias of the extracted 2-D direction increases, the final estimated 3-D absolute attitude error of the capsule also increases. When the bias is between $-5°$ and $5°$, the attitude error estimated by the proposed method is within the acceptable range (about $5°$).

To analyze the effect of noise on the proposed algorithm, we add different noises to the simulated echoes and analyze the performance of the proposed method under different SNR conditions. It is worth mentioning that, since the FEKO simulation is performed under the assumption of a typical turntable model, we need to add the translational motion of the satellite to the simulation data. Here, we still use the STK to obtain the tracking data of the satellite, including the radar observation distance, the azimuth angle, and the elevation angle, as shown in Fig. 15. First, the translational motion is added into the FEKO simulation echoes to generate the raw echoes. Then, noise is added into the raw echoes to obtain the raw echoes with different SNRs. Finally, the ISAR images are obtained by the conventional RD algorithm, including the translational motion compensation [39] and autofocus processing [40]. Fig. 16 shows the error curves of the satellite attitude and size estimation under different SNR conditions. It can be seen that the lower the SNR, the larger the estimated attitude and size error. When the SNR is lower than 10 dB, the attitude estimation error of the proposed method will be larger than $5°$, and the length error will be larger than two meters due to the influence of strong noise, at which point the proposed method will fail. Overall, the proposed algorithm has relatively good noise robustness.

D. Comparison With the Factorization Method

This section shows the advantage of the proposed method compared with the traditional factorization method [21]. The attitude and length of the satellite's capsule will be estimated by measuring the attitude and length of the reconstructed 3-D scatterers. The basic idea of the factorization method is to reconstruct the 3-D scattering point structure of the target by decomposing the trajectory matrix $\mathbf{W}$ via the SVD algorithm. The trajectory matrix $\mathbf{W}$ is obtained by associating scatterers from the ISAR images sequence, which can be expressed as

$$\mathbf{W} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1Q} \\ \vdots & \vdots & \ddots & \vdots \\ x_{F,1} & x_{F,2} & \cdots & x_{F,Q} \\ y_{1,1} & y_{1,2} & \cdots & y_{1,Q} \\ \vdots & \vdots & \ddots & \vdots \\ y_{F,1} & y_{F,1} & \cdots & y_{F,Q} \end{bmatrix} \tag{32}$$

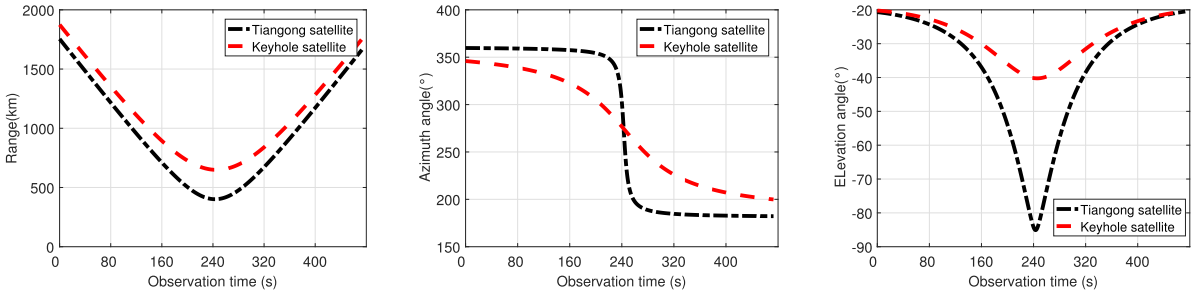
where $(x_{i,j}, y_{i,j})$ is the jth scatterer in the ith frame of ISAR image, $i = 1, 2, \ldots, F$, F is the total frame number of ISAR images, $j = 1, 2, \ldots, Q$, and Q is the total number

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TABLE V

COMPUTATION TIME OF ONE ITERATION AND CONVERGENCE FOR PSO AND THE PROPOSED METHOD

	Methods	TianGong (TG)	KeyHole (KH)
Computation time for one iteration [s]	Proposed method	0.019	
	PSO	0.017	
Number of iterations for convergence	Proposed method	17	27
	PSO	103	141
Computation time for convergence [s]	Proposed method	0.323	0.513
	PSO	1.751	2.397



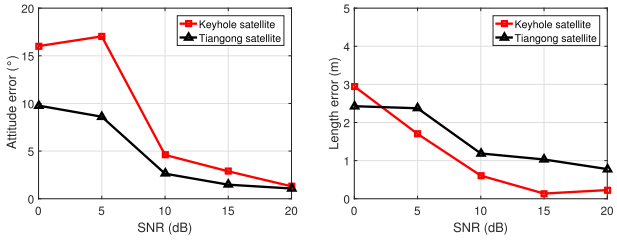
(a) (b) (c)

Fig. 15. (a) Distance, (b) azimuth angle, and (c) elevation angle of the satellite in orbit acquired using STK software.

TABLE VI

ESTIMATED ATTITUDES AND LENGTHS OF TWO SATELLITES BY THE PROPOSED METHOD AND THE FACTORIZATION METHOD

Target		KeyHole (KH)		TianGong (TG)	
		Attitude	Length	Attitude	Length
True values		(0.500,0,0.866)	12 [m]	(0.163,-0.925,-0.342)	11.488 [m]
Proposed method		(0.499,0.005 ,0.867)	12.405 [m]	(0.164,-0.927,-0.337)	12.260 [m]
Factorization method		Unknown	12.996 [m]	Unknown	10.498 [m]
Errors	Proposed method	0.493°	0.405 [m]	0.465°	0.771 [m]
	Factorization method	Unknown	0.996 [m]	Unknown	0.990 [m]



(a) (b)

Fig. 16. Curves of attitude and size estimation errors with SNR. (a) Variation curve of attitude error with SNR. (b) Variation curve of size error with SNR.

of scatterers. By decomposing $\mathbf{W}$ via SVD, we can get

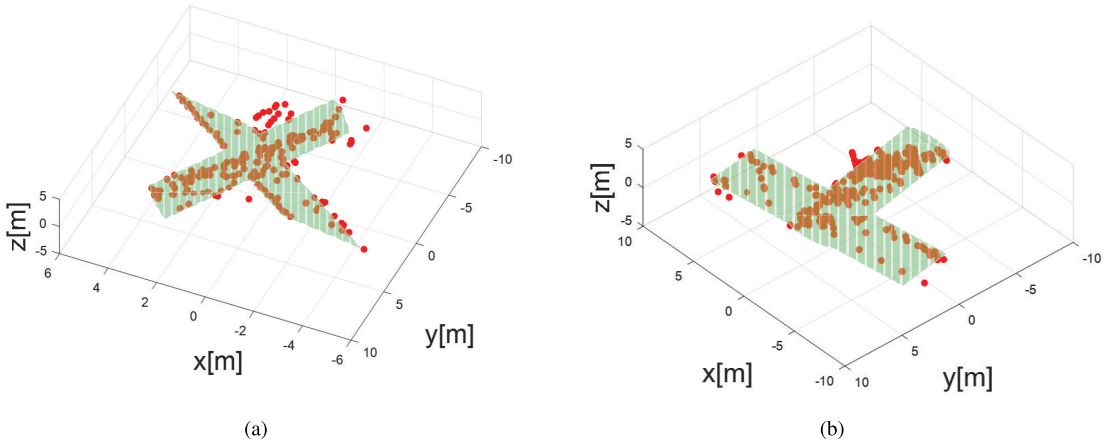
$$\mathbf{W} = \mathbf{U}\Sigma\mathbf{V} = \left(\mathbf{U}\sqrt{\Sigma}\right)\left(\sqrt{\Sigma}\mathbf{V}\right) = \mathbf{MP}. \tag{33}$$

The decomposing of (33) is not unique, for any $\mathbf{CC}^{-1} = \mathbf{I}$, $\mathbf{MP} = (\mathbf{MC})(\mathbf{C}^{-1}\mathbf{P})$. Under the constrain of (34), $\hat{\mathbf{M}} = \mathbf{MC}$, $\hat{\mathbf{P}} = \mathbf{C}^{-1}\mathbf{P}$ can be one acceptable solution, where $\hat{\mathbf{M}} = [\mathbf{i}_1, \mathbf{i}_2, \ldots, \mathbf{i}_F, \mathbf{j}_1, \mathbf{j}_2, \ldots, \mathbf{j}_F]^T$. $\hat{\mathbf{P}}$ is the 3-D scatterers of target. After getting $\hat{\mathbf{P}}$, the satellite capsule's length can be obtained with some manual effort by measuring the distance of the ends points of the satellite's capsule. Fig. 17 gives the reconstruct results of KH and TG, from which we can see the target's rough structure and shape

$$\begin{cases} \|\mathbf{i}_k\| = 1 \\ \|\mathbf{j}_k\| = 1 \\ \mathbf{i}_k \cdot \mathbf{j}_k = 0. \end{cases} \tag{34}$$

Table VI gives the estimated capsule lengths and attitudes. We can see that we cannot get the absolute attitude of the

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(a) (b)

Fig. 17. Reconstruct results of these two satellite. (a) 3-D scatterers of KH. (b) 3-D scatterers of TG.

satellite's capsule as the decomposing of the trajectory matrix $\mathbf{W}$ is a multisolution problem. For any matrix $\mathbf{C}\mathbf{C}^T = \mathbf{I}$, (35) is another solution to the decomposing

$$\mathbf{W} = \tilde{\mathbf{M}}\tilde{\mathbf{P}} = \left(\hat{\mathbf{M}}\mathbf{C}\right)\left(\mathbf{C}^T\hat{\mathbf{P}}\right). \tag{35}$$

The matrix $\mathbf{C}$ leads to an unknown rotation operation between the estimated 3-D scatterers matrix $\hat{\mathbf{P}}$ and the absolute solution $\tilde{\mathbf{P}}$, making it impossible to estimate the absolute attitude of the satellite's capsule from matrix $\hat{\mathbf{P}}$. Table VI shows that the errors of estimated lengths by factorization method are slightly larger than those estimated by the proposed method. Therefore, we can conclude that the proposed method has a better performance at estimating the attitude and the length of the satellite's capsule than the factorization method.

V. Conclusion

This article proposes a novel method to estimate satellite absolute attitude and size from the ISAR images. The proposed method bridged RD images and the target's motion state, accommodating the ISAR geometric projection model. By treating the satellite's capsules as individual structures and segmenting them from the ISAR images via Pix2pixGAN, the 2-D features of the satellite's capsules can be obtained by employing PCA. These extracted features in the observation images recovered the target attitude and size through a proposed parametric optimization, which was solved by the gradient-based iteration method. Experiments based on the electromagnetic simulation data illustrated the advantage of the proposed algorithm in attitude and size estimation. Compared with the factorization-based reconstruction method, the proposed algorithm can estimate the absolute attitude and size of the satellite simultaneously with higher accuracy of size estimation.

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